

Categories, Proofs, and Processes:

Lecture 17

Wira Azmoon Ahmad

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Adjunctions

Unit

$$\mathcal{C} \begin{array}{c} \xrightarrow{F} \\ \perp \\ \xleftarrow{G} \end{array} \mathcal{D}$$

is an adjunction iff $\forall X \in \text{Ob}(\mathcal{C})$, there exists a unit

$$\eta_X : X \rightarrow GF(X)$$

such that

$$\forall f : X \rightarrow G(Y), \exists ! \hat{f} : F(X) \rightarrow Y$$

where

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & GF(X) \\ & \searrow f & \downarrow G(\hat{f}) \\ & & G(Y) \end{array}$$

Example 1.1. Makes more sense as unit, unit version.

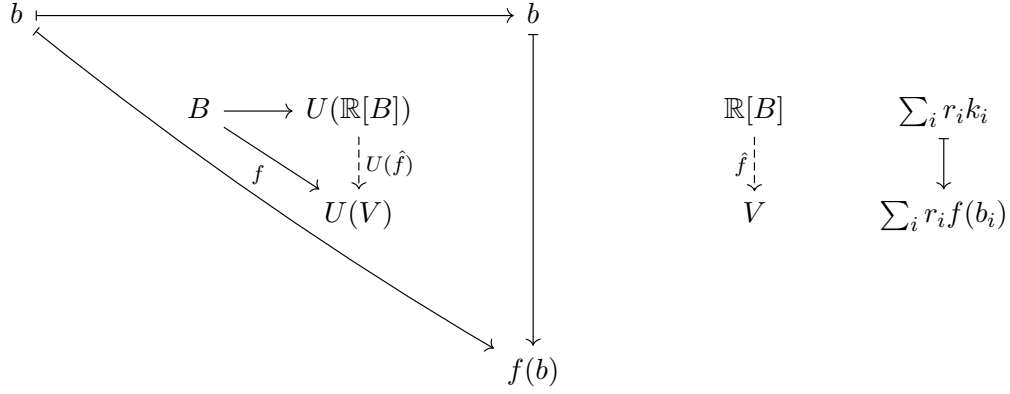
$$\mathbf{Set} \begin{array}{c} \xrightarrow{\mathbb{R}[-]} \\ \perp \\ \xleftarrow{U} \end{array} \mathbf{Vect}_{\mathbb{R}}$$

$\mathbb{R}[X]$ vector space spanned by X .

$$v \in \mathbb{R}[X], v = \sum_i r_i x_i, \quad r_i \in \mathbb{R}, x_i \in X$$

$$\mathbb{R}[f] : \mathbb{R}[X] \rightarrow \mathbb{R}[Y]$$

$$\mathbb{R}[f] \left(\sum_i r_i x_i \right) = \sum_i r_i f(x_i)$$



Example 1.2. Makes more sense as co-unit, unit version.

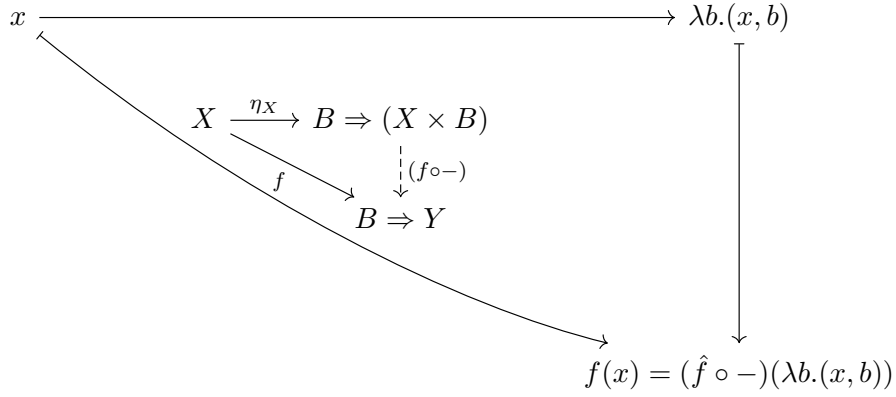
$$(- \times B) : \mathbf{Set} \rightarrow \mathbf{Set}$$

$$\begin{cases} A \mapsto A \times B \\ f \mapsto f \times 1_B \end{cases}$$

$$g : X \rightarrow Y$$

$$B \Rightarrow g : B \Rightarrow X \rightarrow B \Rightarrow Y$$

$$(B \Rightarrow g)(h) := g \circ h$$



$$\hat{f} : X \times B \rightarrow Y$$

$$\hat{f}(x, b) = f(x)(b)$$

$$(\hat{f} \circ -)(\lambda b.(x, b)) = \lambda b.\hat{f}(x, b) = \lambda b.f(x)(b) = f(x)$$

Last equality by η expansion.

Notes

- Example: X is set, $GF(X)$ is a list.
- More useful for the left adjoints.

Bijection

$$\mathcal{D}(F(X), Y) \cong \mathcal{C}(X, G(Y))$$

Co-Unit

$$\begin{array}{ccc} & F & \\ \mathcal{C} & \xrightarrow{\quad} & \mathcal{D} \\ & \perp & \\ & G & \end{array}$$

is an adjunction iff $\forall X \in \text{Ob}(\mathcal{C})$, there exists a co-unit

$$\varepsilon_Y : FG(Y) \rightarrow Y$$

such that

$$\forall f : F(X) \rightarrow Y, \exists ! \tilde{f} : X \rightarrow G(Y)$$

where

$$\begin{array}{ccc} F(X) & & \\ F(\tilde{f}) \downarrow & \searrow f & \\ FG(Y) & \xrightarrow{\varepsilon_Y} & Y \end{array}$$

Notes

Co-unit is some kind of evaluation.

Example 1.3. Makes more sense as unit, co-unit version.

$$F = \mathbb{R}[-], G = U$$

$$\begin{array}{ccc} & \mathbb{R}[-] & \\ \mathbf{Set} & \xrightarrow{\quad} & \mathbf{Vect}_{\mathbb{R}} \\ & \perp & \\ & U & \end{array}$$

"Forget that Y is a vector space, treat it as a set"

$$\begin{array}{ccc} \mathbb{R}[X] & & \\ F(\tilde{f}) \downarrow & \searrow f & \\ \mathbb{R}[U(Y)] & \xrightarrow{\varepsilon_Y} & Y \\ \uparrow \wr & & \\ \sum_i r_i \cdot [v_i] & \longmapsto & \sum_i r_i v_i \end{array}$$

$$\tilde{f}(x) = f(x)$$

Example 1.4. Makes more sense as co-unit.

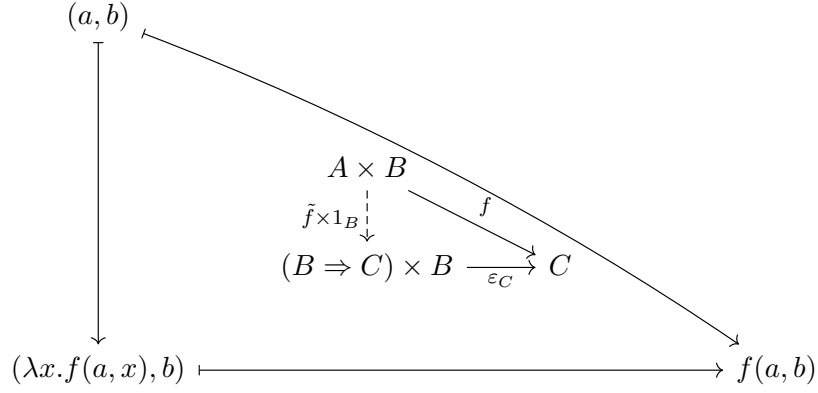
$$(- \times B) : \mathbf{Set} \rightarrow \mathbf{Set}$$

$$\begin{cases} A \mapsto A \times B \\ f \mapsto f \times 1_B \end{cases}$$

$$\mathbf{Set}(A \times B, C) \cong \mathbf{Set}(A, B \Rightarrow C)$$

$$\begin{array}{ccc} & \xrightarrow{(- \times B)} & \\ \mathbf{Set} & \perp & \mathbf{Set} \\ & \xleftarrow{G=(B \Rightarrow -)} & \end{array}$$

$$FG(C) = (B \Rightarrow C) \times B$$



$$\tilde{f} : A \rightarrow (B \Rightarrow C)$$

$$\tilde{f} :: a \mapsto \lambda x. f(a, x)$$

$$\tilde{f} :: a \mapsto f(a, -)$$

Notation mapping of the same thing.